

B-ES2-C6 MULTIPROTOCOL ROOM SENSOR

24 V / IDM interface and protected 5 V supply

ESP32-C6, USB-C and local sensing

0-10 V and isolated RS-485

PCB layout for the B-ES2-C6 Multiprotocol Room Sensor. The layout is divided into three main functional blocks: 24V / IDM interface and protected 5V supply (left), ESP32-C6, USB-C and local sensing (center), and 0-10 V and isolated RS-485 (right). Key components include a USB-C connector (J1), a USB-C to UART module (U1), an ESP32-C6 module (U2), a USB-C to RS-485 module (U3), a 0-10 V sensor module (U4), and a 0-10 V sensor module (U5). The layout includes various passive components like resistors (R1-R10), capacitors (C1-C17), and a ferrite bead (FB1). The board is populated with components like the LMR36100 (U1), AP2112K-3.3TRG1 (U2), MCP4725A0T-E/CH (U3), and OP497IDBVR (U4). The layout also shows the placement of test points (TP1-TP12) and a status LED (LED1). The board is labeled 'B-ES2-C6 MULTIPROTOCOL ROOM SENSOR' and 'IDM SmartCloud Platform'.

File: IDM-RoomSensor-ESP-B-ES2-C6.kicad_sch
Title: IDM Room Sensor B-ES2-C6 - USB-C / RS-485 / 0-10 V

Size: A4 Date: 2026-08-07 Rev: B-ES2-C6